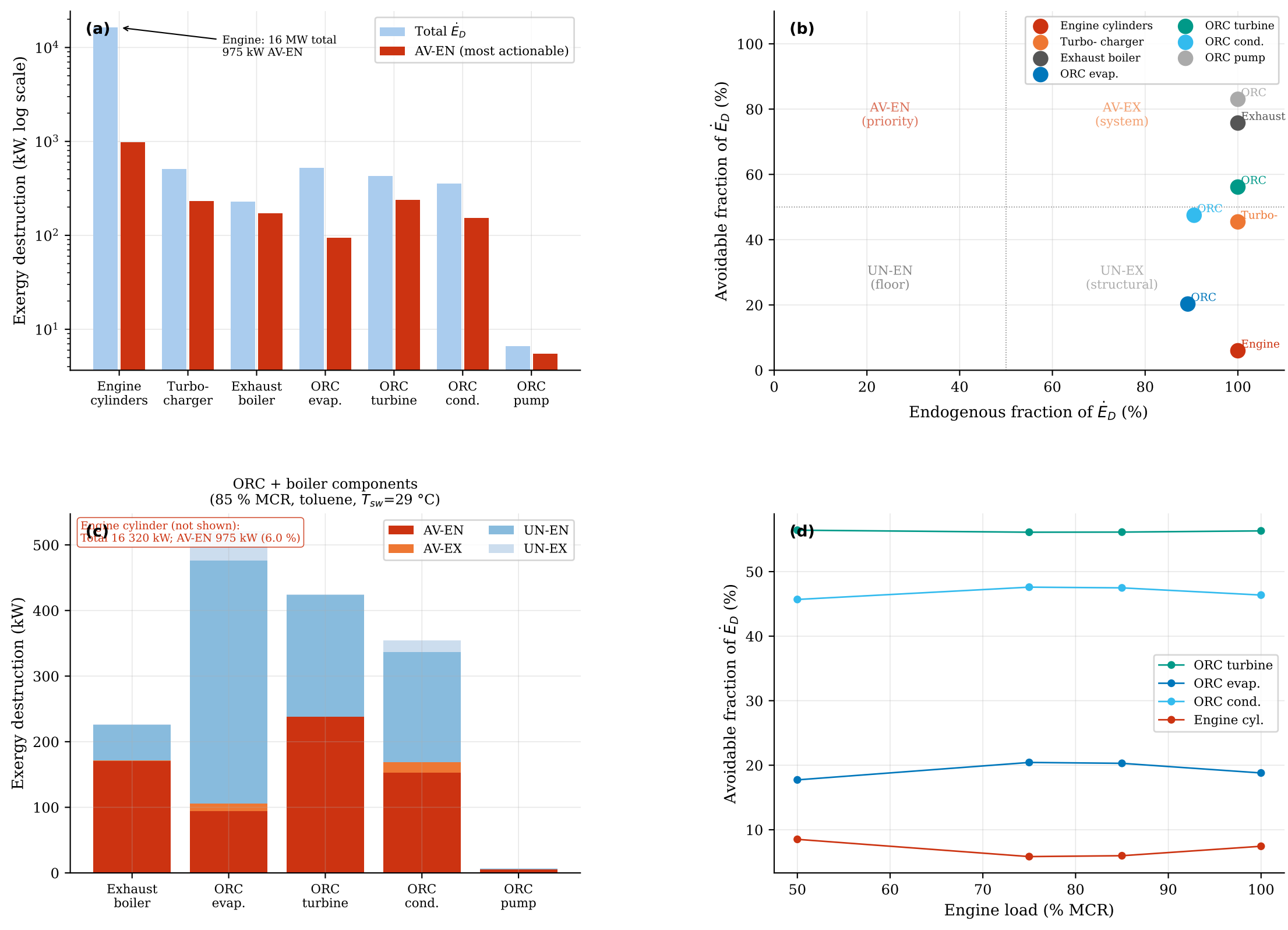
(a)
Engine: 16 MW total
975 kW AV-EN
Total $\dot{E}_D$
AV-EN (most actionable)
Exergy destruction (kW, log scale)
Engine cylinders
Turbo-charger
Exhaust boiler
ORC evap.
ORC turbine
ORC cond.
ORC pump
(b)
Engine cylinders
Turbo- charger
Exhaust boiler
ORC evap.
ORC turbine
ORC cond.
ORC pump
AV-EN (priority)
AV-EX (system)
UN-EN (floor)
UN-EX (structural)
Avoidable fraction of $\dot{E}_D$ (%)
Endogenous fraction of $\dot{E}_D$ (%)
(c)
ORC + boiler components
(85 % MCR, toluene, T_{sw}=29 °C)
Engine cylinder (not shown):
Total 16 320 kW; AV-EN 975 kW (6.0 %)
AV-EN
AV-EX
UN-EN
UN-EX
Exergy destruction (kW)
Exhaust boiler
ORC evap.
ORC turbine
ORC cond.
ORC pump
(d)
ORC turbine
ORC evap.
ORC cond.
Engine cyl.
Avoidable fraction of $\dot{E}_D$ (%)
Engine load (% MCR)